

Credit Card Approval Prediction Using Machine Learning

Project Description:

The Credit Card Approval Prediction system is a Machine Learning-based web application that predicts whether a customer's credit card application is likely to be approved or rejected based on personal and financial information. The application uses a trained Random Forest Classifier to analyze user inputs and estimate the approval probability.

The system provides the prediction result along with the approval/rejection probability and assigns a risk category (Low, Medium, or High). Every prediction is stored in a SQLite database, allowing users to review previous predictions and visualize approval statistics through an interactive dashboard.

The application is developed using Flask, HTML, CSS, SQLite, Joblib, Pandas, and Matplotlib, providing a complete end-to-end Machine Learning solution.

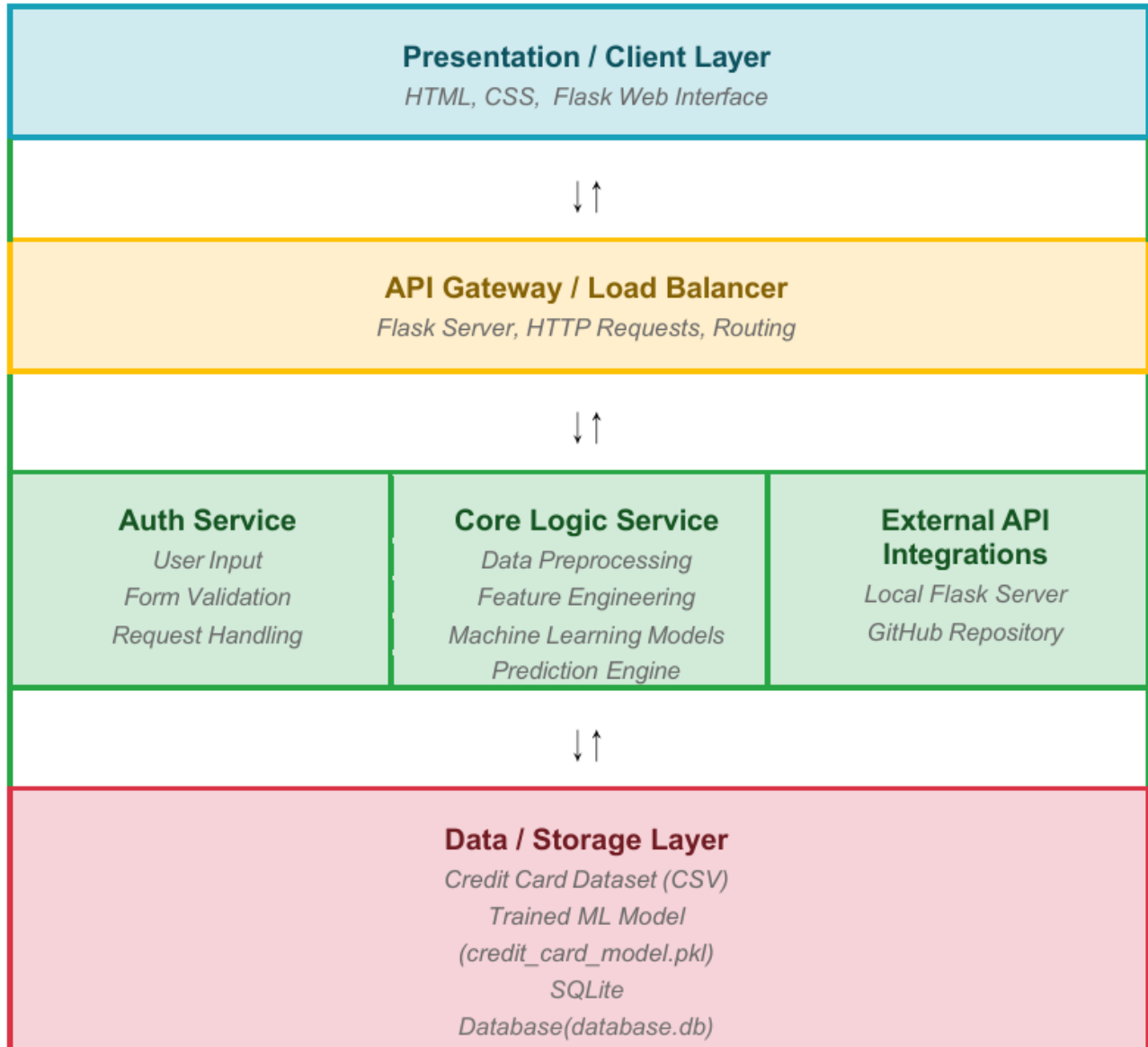
Scenarios

Scenario 1: A salaried employee with a high annual income, higher education, stable employment, and good family profile applies for a credit card. The system predicts Approved with Low Risk.

Scenario 2: A customer with average income and limited work experience applies for a credit card. The system predicts Approved with Medium Risk.

Scenario 3: A customer with low income and unstable employment applies for a credit card. The system predicts Rejected with High Risk.

Technical Architecture:



Description: CaptionAI utilizes a modular three-tier architecture to deliver rapid, AI-driven content generation. The frontend provides a responsive user interface built with HTML and CSS while the Flask-based backend orchestrates data validation and prompt construction. It interfaces with the Groq API using the LLaMA 3.3-70B model to generate tailored social media content, with secure deployment managed locally via Flask and exposed publicly through Ngrok.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

Pre-requisites:

- Python 3.12+
- Flask
- Pandas
- NumPy
- Scikit-learn
- Joblib
- SQLite3
- Matplotlib
- VS Code
- Git & GitHub

Project Workflow:

Milestone 1

- Dataset Collection
- Requirement Analysis
- Feature Selection

Milestone 2

- Data Cleaning
- Data Preprocessing
- Model Training

Milestone 3

- Flask Application Development
- HTML Form Design
- Prediction Logic

Milestone 4

- Database Integration
- Dashboard Development
- History Page

Milestone 5

- Testing
- Documentation
- GitHub Upload

Milestone 1

Dataset Collection:

Dataset consists of applicant information including

- Gender
- Income
- Education
- Occupation
- Family Status
- Housing Type
- Days Birth
- Days Employed
- Family Members

The dataset is cleaned and encoded before training.

Development Environment :

Software Used

- Python
- VS Code
- Flask
- SQLite
- GitHub

```
def home():  
    return render_template("index.html")  
  
@app.route("/predict", methods=["POST"])  
def predict():  
    age = int(request.form["age"])  
    experience = int(request.form["experience"])  
  
    # Convert Age & Experience into dataset format  
    days_b = int(request.form["days_birth"]) * 365  
    days_employed = int(request.form["days_employed"]) * 365  
  
    values = [[  
        int(request.form["gender"]),  
        int(request.form["car"]),  
        int(request.form["realty"]),  
        int(request.form["children"]),  
        float(request.form["income"]),  
        int(request.form["income_type"]),  
        int(request.form["education"]),  
        int(request.form["family"]),  
        int(request.form["housing"]),  
        days_b,   
        days_employed,   
        int(request.form["mobile"]),  
        0, # Work Phone  
        0, # Phone  
        int(request.form["email"]),  
        int(request.form["occupation"]),  
        float(request.form["family_members"])  
    ]]  
  
    probability = model.predict_proba(values)[0]  
    approval_probability = round(probability[0] * 100, 2)
```

Milestone 2:

Core Functionalities Development

- **Milestone 2**
- **Data Preprocessing**
- **Performed**
- **Missing Value Handling**
- **Label Encoding**
- **Feature Selection**
- **Data Type Conversion**
- **Prepared the dataset for Random Forest training.**
- **Model Training**
- **Algorithm Used**
- **Random Forest Classifier**
- **Model saved as**
- **credit_card_model.pkl** as fallback strategies.

Milestone 3: Flask Application Development Implemented Routes

/

Home Page

/predict

Prediction

/history

Prediction History

/dashboard

Dashboard

/download

Download Prediction History

Database

SQLite Database

Table

Approval_Prediction

Stores

Applicant ID

Prediction Result

Risk Category

Prediction Date

Milestone 4:

Frontend Development :

- Developed using
- HTML
- CSS
- Features
- User-friendly Input Form
- Prediction Result
- Risk Category
- Navigation Buttons
- Responsive Layout
- Dashboard
- Dashboard displays
- Total Applications
- Approved Applications
- Rejected Applications
- Approval Rate
- Pie Chart
- Bar Chart

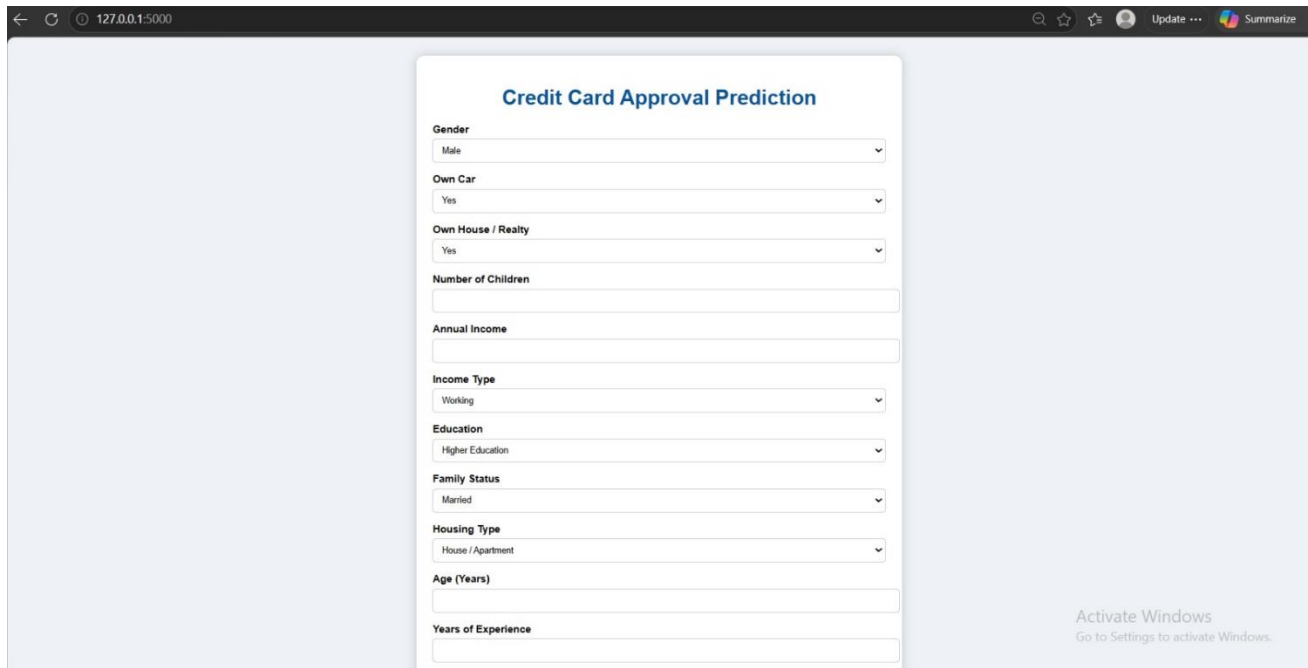
Milestone 5:

Frontend Development

- **Developed using**
 - HTML
 - CSS
- **Features**
 - User-friendly Input Form
 - Prediction Result
 - Risk Category
 - Navigation Buttons
 - Responsive Layout
 - Dashboard
 - Dashboard displays
 - Total Applications
 - Approved Applications
 - Rejected Applications
 - Approval Rate
 - Pie Chart
 - Bar Chart

Exploring the Website's Web Pages:

Home / Generator Page:



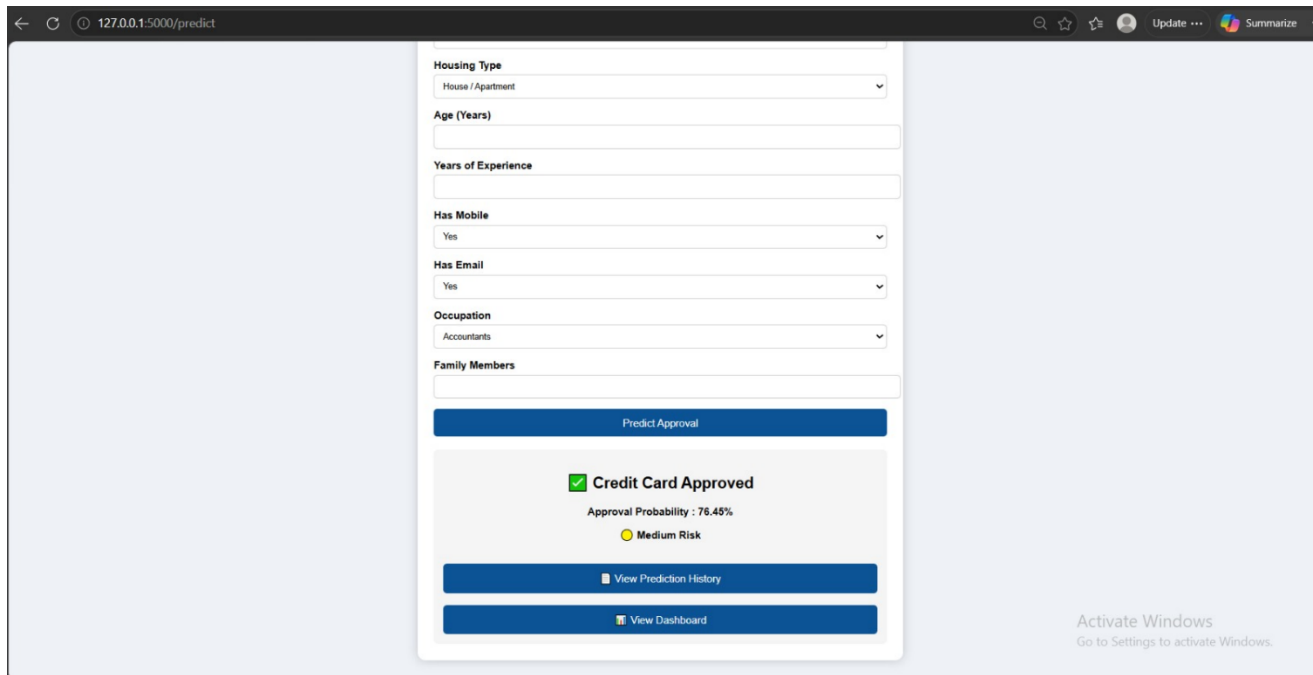
The screenshot shows a web browser window displaying a form titled "Credit Card Approval Prediction". The form is centered on a light blue background. The browser's address bar shows "127.0.0.1:5000". The form contains the following fields:

- Gender:** A dropdown menu with "Male" selected.
- Own Car:** A dropdown menu with "Yes" selected.
- Own House / Realty:** A dropdown menu with "Yes" selected.
- Number of Children:** A text input field.
- Annual Income:** A text input field.
- Income Type:** A dropdown menu with "Working" selected.
- Education:** A dropdown menu with "Higher Education" selected.
- Family Status:** A dropdown menu with "Married" selected.
- Housing Type:** A dropdown menu with "House / Apartment" selected.
- Age (Years):** A text input field.
- Years of Experience:** A text input field.

At the bottom right of the form, there is a watermark that reads "Activate Windows Go to Settings to activate Windows."

Description: The Home Page allows users to enter applicant details such as gender, income, education, occupation, family members, experience, and housing type. After submitting the form, the machine learning model predicts the approval status.

Results Section:



127.0.0.1:5000/predict

Housing Type
House / Apartment

Age (Years)

Years of Experience

Has Mobile
Yes

Has Email
Yes

Occupation
Accountants

Family Members

Predict Approval

 **Credit Card Approved**
 Approval Probability : 76.45%
 Medium Risk

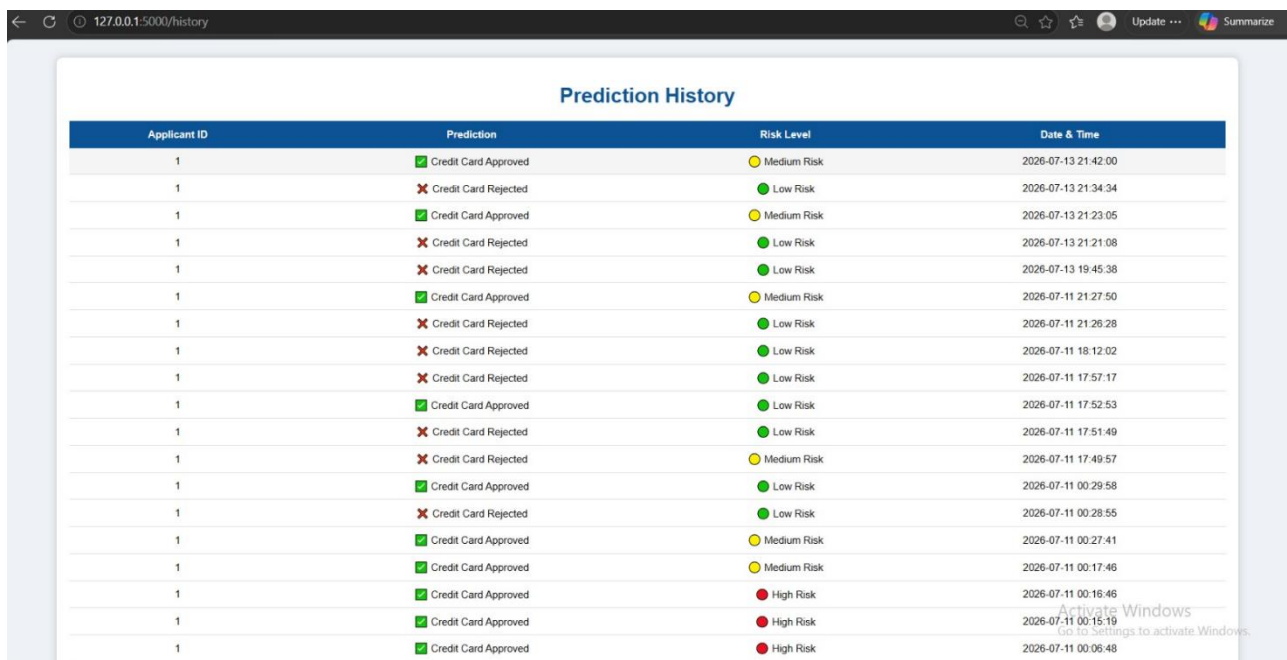
View Prediction History

View Dashboard

Activate Windows
Go to Settings to activate Windows.

Description: Displays
 Approval/Rejection Result
 Prediction Probability
 Risk Category

Prediction History :



127.0.0.1:5000/history

Prediction History

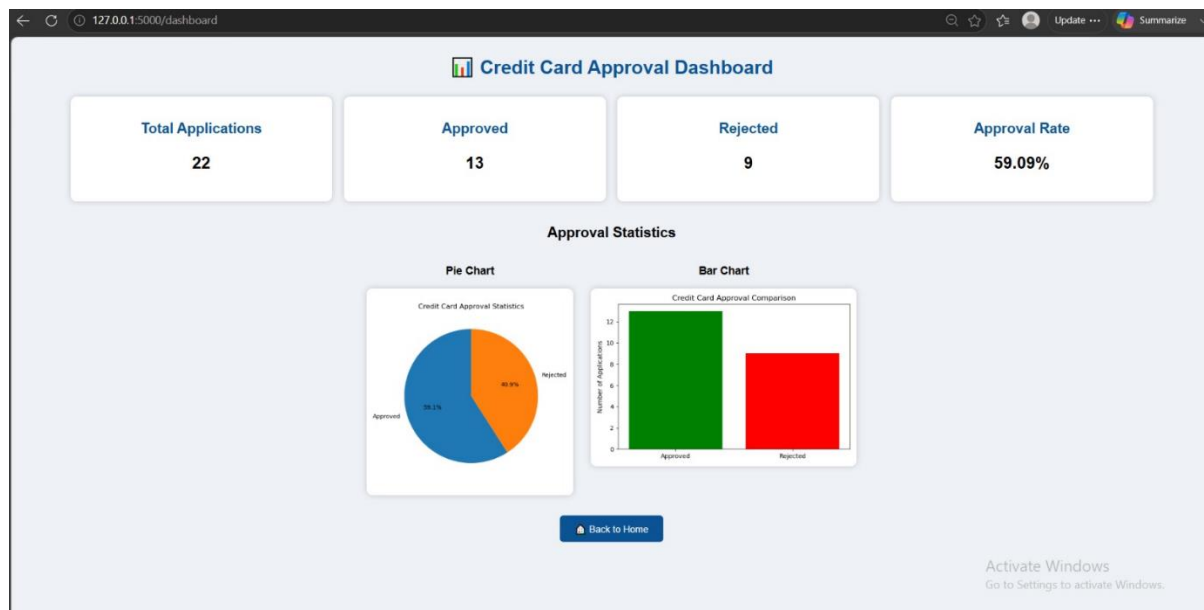
Applicant ID	Prediction	Risk Level	Date & Time
1	 Credit Card Approved	 Medium Risk	2026-07-13 21:42:00
1	 Credit Card Rejected	 Low Risk	2026-07-13 21:34:34
1	 Credit Card Approved	 Medium Risk	2026-07-13 21:23:05
1	 Credit Card Rejected	 Low Risk	2026-07-13 21:21:08
1	 Credit Card Rejected	 Low Risk	2026-07-13 19:45:38
1	 Credit Card Approved	 Medium Risk	2026-07-11 21:27:50
1	 Credit Card Rejected	 Low Risk	2026-07-11 21:26:28
1	 Credit Card Rejected	 Low Risk	2026-07-11 18:12:02
1	 Credit Card Rejected	 Low Risk	2026-07-11 17:57:17
1	 Credit Card Approved	 Low Risk	2026-07-11 17:52:53
1	 Credit Card Rejected	 Low Risk	2026-07-11 17:51:49
1	 Credit Card Rejected	 Medium Risk	2026-07-11 17:49:57
1	 Credit Card Approved	 Low Risk	2026-07-11 00:29:58
1	 Credit Card Rejected	 Low Risk	2026-07-11 00:28:55
1	 Credit Card Approved	 Medium Risk	2026-07-11 00:27:41
1	 Credit Card Approved	 Medium Risk	2026-07-11 00:17:46
1	 Credit Card Approved	 High Risk	2026-07-11 00:16:46
1	 Credit Card Approved	 High Risk	2026-07-11 00:15:19
1	 Credit Card Approved	 High Risk	2026-07-11 00:06:48

Activate Windows
Go to Settings to activate Windows.

Description :

Displays all previous predictions stored in SQLite along with prediction date and risk category.

Dashboard:



Description :

- Displays
- Total Applications
- Approved Applications
- Rejected Applications
- Approval Percentage
- Pie Chart
- Bar Chart

Conclusion:

The Credit Card Approval Prediction project successfully demonstrates the application of Machine Learning in financial decision-making. A Random Forest Classifier was trained using applicant information and integrated into a Flask web application to predict credit card approval. The application also stores prediction history in a SQLite database and provides graphical insights through an interactive dashboard.

This project improves decision-making by reducing manual effort and providing fast, accurate predictions. Future enhancements include user authentication, cloud deployment, improved model accuracy using advanced algorithms, email notifications, and real-time credit score integration.